NoSQL Services
Amazon DynamoDB
Tutorial

Topics

1. Using Amazon DynamoDB via the AWS SDK for Python

Prerequisite

- Complete this tutorial using the AWS Academy Learner Lab dedicated for our module[1].
- Note that if you are completing this tutorial using the AWS Academy Learner Lab, there is a restriction on the **AWS Regions** that can be used, namely all service access is limited to the **us-east-1** and **us-west-2** AWS Regions unless mentioned otherwise.
- This tutorial assumes that you are using the AWS Cloud9 Environment set up for developing in Python. Otherwise please complete the section *Setting up Cloud9 for Cloud Application Development in Python* of the Tutorial on AWS Cloud9 for Python.

Working with Amazon DynamoDB via the AWS SDK for Python

Amazon DynamoDB[2] is a key-value and document database that delivers single-digit millisecond performance at any scale, and provides a durable databases with built-in security.

The aim of this tutorial is to learn how to programmatically use Amazon DynamoDB. More specifically, at the end of this tutorial you should know to perform the following tasks:

[1] This assumes that you have accepted the invitation to join the AWS Academy Learner Lab received via your NCI email account.

[2] https://aws.amazon.com/dynamodb/

- Create a DynamoDB table
- Store data in a DynamoDB table
- Retrieve the information/attributes of an item
- Update an item in a table
- Delete an item from a table
- Query a table to find specific information
- Delete a table

Recall that the AWS SDKs provide two levels of APIs for the AWS services, namely Client (or "low-level") APIs and Resource APIs. Please consult the official boto3 DynamoDB documentation available at https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/dynamodb.html.

The documentation of the boto3 DynamoDB Client, which is a low-level API representing the Amazon DynamoDB service, is available at https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/dynamodb.html#client, and the documentation of the boto3 DynamoDB service resource is available at https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/dynamodb.html#service-resource.

Sample Python Program On the module's Moodle page is available a sample program in the file named *dynamodb_ demo.py* that shows examples on how to perform operations with Amazon DynamoDB programmatically via the available SDK. Explanations are provided during class.

Running the sample Python program

To run the sample Python program *dynamodb_ demo.py*, at the terminal run the following command

```
python dynamodb_demo.py
```

Exercises

Using the previous examples as an introduction to working with Amazon DynamoDB via the available SDK, provide an implementation for the following tasks:

1. Update an item in a table
2. Delete an item from a table
3. Query a table to find specific information
    - Explore the different methods available to perform such an operation
4. Delete a table

Resources

- Amazon DynamoDB https://aws.amazon.com/dynamodb/
- Boto3 Amazon DynamoDB Documentation https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/dynamodb.html
- Boto3 Amazon DynamoDB Client https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/dynamodb.html#client
- Boto3 Amazon DynamoDB Service Resource https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/dynamodb.html#service-resource
- Getting Started Developing with Python and DynamoDB https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/GettingStarted.Python.html
- AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/welcome.html
- Python Tutorial for AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/sample-python.html